# Yaxin Du

Shanghai, China   dorothydu@sjtu.edu.cn   +86 132-1905-8049   Google Scholar   Github

## Education

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| **Ph.D.** | **Shanghai Jiao Tong University**, Information and Communication Engineering. (Supervisor: Siheng Chen) | Sept 2022 – Present |
| **B.Eng** | **University of Electronic Science and Technology of China**, Communication Engineering | Sept 2018 – June 2022 |
| **B.Eng** | **University of Glasgow**, Communication Engineering | Sept 2018 – June 2022 |

## Research Interests

**Code Generation**: Building automatic & reliable code generation systems.
**Multi-Agent System**: Building self-evolving, LLM-driven collaboration frameworks for various complex tasks.
**Federated Learning**: Exploring privacy-preserving federated learning for LLMs.

## Research Experience

**Agent Tool-using Data Synthesis** June 2024 – Oct 2025

- Proposed **InfoMosaic-Bench**, a multi-source information seeking task for tool-augmented agents. Developed a scalable synthesis pipeline that emphasized compositional retrieval and grounding.
- Built **MCP-Flow** a large-scale Tool Selection Dataset, capturing *selection, ordering, and parameterization* signals with executable traces; enabled both SFT and RL on agentic tool use.
- Synthesized 3K high-quality websearch QA pairs and trajectories for **BrowseMaster**; resulting an 8B model to 21% on BrowseComp.

**Repo-level Software Development Training Dataset** Oct 2024 – May 2025

- Curated **SWE-Dev**, the first large-scale benchmark for autonomous, repo-level feature development (**14k** train / **500** test tasks over **1k+** repos). Each sample is coupled with full project context, unit tests, Docker runtime, and execution harness—enabling code-running supervision.
- Trained SWE-Dev dataset across **single-agent** LLM supervised fine-tuning, reinforcement learning, and **multi-agent** finetuning, verifing the data quality.
- Open-sourced the full dataset, training recipes, and inference/evaluation toolkit on GitHub, with 50+ stars and benchmarked by **Kimi-K2** on the 0907-version release.

**Multi-agent System for Software Development** Jul 2024 – Oct 2024

- Developed EvoMAC, a self-evolving multi-agent system for coding, which is the top-2 highest-scoring multi-agent paper in ICLR 2025
- Designed and developed rSDE-Bench, a requirement-oriented benchmark for evaluating software-level development on website and game. All tests are blackboxed

**Federated Learning for Large Language Models** May 2023 – Jul 2024

- Proposed **FedDQC**, a novel data quality control framework that enhances federated instruction tuning by integrating instruction-response alignment scoring, single-shot noisy data filtering, and hierarchical tuning.
- Built **OpenFedLLM**, a concise and integrated framework supporting federated instruction tuning, value alignment, and 7 representative FL algorithms, enabling collaborative LLM training on distributed private data.
- Developed **FedLLM-Bench**, the first realistic benchmark for FedLLM, featuring 8 training methods, 4 realistic datasets, and 6 evaluation metrics to capture diverse real-world scenarios, facilitating fair comparison and accelerating FedLLM research.

**Behavior Classification Based on Cyclostationarity of Radar Signals** Feb 2020 – Apr 2022

- Led student research team on vital sign detection and activity classification under noise and interference.
- **EMSIG Challenge:** First place, applying Vision Transformer on Doppler features detection.

- **CIE Radar 2021:** Cyclostationary feature extraction with feature-level fusion.
- **Radar Conf 2022:** Applying ViT into cyclostationary analysis and reached SOTA performance.
- **BS Thesis:** Comparison of Doppler and cyclostationary methods using ViT and TST. Top 1% best BS Thesis in UESTC.

## Project

**AI Study Buddy & Research Collaboration Platform** Dec 2024 – Feb 2025

- Led the agent training pipeline: curated 100 K research dialogues and fine-tuned a 72B LLM with SFT and DPO, boosting single-turn answer accuracy by 29%.
- Implemented tool-calling in chatbot: integrated web & paper search with query latency < 800 ms, answer factuality +35%.
- Project scheduled for university-wide launch on the SJTU smart campus platform in September.

**AI Scientist Researcher** Mar 2025 – Aug 2025

- Led the dataset team: collected 1.33B high-quality data from undergraduate-level and graduate-level textbooks for training (oriented HLE-Bench).
- Curate multimodal document understanding and RAG system for scientific research, already used by the Yazhou Bay National Laboratory.

## Internship

**Research Intern, TikTok AI Innovation Center** (supervised by Qian Liu) Sept 2025 – Present

- Trained **co-evolving code agents** through reinforcement learning, where Coder and Verifier collaboratively improve via iterative code–test interaction.
- Implemented BoN-based evaluation and alternating optimization, achieving a **+5.77%** performance gain on CodeContest compared to single-agent training.

## Selected Publications

**InfoMosaic-Bench: Evaluating Multi-Source Information Seeking in Tool-Augmented Agents** (Under Review) Sept 2025
***Yaxin Du***, Yuanshuo Zhang, Xiyuan Yang, Yifan Zhou, Cheng Wang, Gongyi Zou, Xianghe Pang, Zhiyu Li, Siheng Chen

**SWE-Dev: Evaluating and Training Autonomous Feature-Driven Software Development** (Under Review) May 2025
***Yaxin Du****, Yuzhu Cai*, Yifan Zhou, Cheng Wang, Yu Qian, Xianghe Pang, Qian Liu, Yue Hu, Siheng Chen

**Data Quality Control in Federated Instruction-tuning of Large Language Models** (ACL 2025 findings) Feb 2025
***Yaxin Du***, Rui Ye, Fengting Yuchi, Wanru Zhao, Jingjing Qu, Yanfeng Wang, Siheng Chen

**Enhancing Data Quality in Federated Fine-tuning of Large Language Models** (ICLR Workshop 2024) May 2024
Wanru Zhao*, ***Yaxin Du****, Nicholas Lane, Siheng Chen, Yanfeng Wang

**A ViT Approach for Short-range Behaviour Recognition Using Radar Signals** (RadarConf workshop 2022) Mar 2022
***Yaxin Du***, Bingliang Li, Jipeng Li, Francesco Fioranelli, Julien Le Kernec

**MAS-GPT: Training LLMs to Build LLM-based Multi-Agent Systems** (ICML 2025) Mar 2025
Rui Ye, Shuo Tang, Rui Ge, ***Yaxin Du***, Zhenfei Yin, Siheng Chen, Jing Shao

**Self-evolving Multi-agent Collaboration Networks for Software Development** (ICLR2025) Dec 2024
Yue Hu, Yuzhu Cai, ***Yaxin Du***, Xinyu Zhu, Xiangrui Liu, Zijie Yu, Yuchen Hou, Shuo Tang, Siheng Chen

**FedLLM-Bench: Realistic Benchmarks for Federated Learning of Large Language Models** (NeurIPS 2024) Sep 2024
Rui Ye, Rui Ge, Xinyu Zhu, Jingyi Chai, ***Yaxin Du***, Yang Liu, Yanfeng Wang, Siheng Chen

**Federated Instruction Tuning of LLMs with Domain Coverage Augmentation** (EMNLP 2025, findings) Sep 2024

Zezhou Wang, ***Yaxin Du***, Xingjun Ma, Yugang Jiang, Zhuzhong Qian, Siheng Chen

**Radar-based Human Activity Recognition using Denoising Techniques to Enhance Classification Accuracy** (IET Radar, Sonar & Navigation 2024) Feb 2024
Ran Yu, ***Yaxin Du***, Jipeng Li, Antonio Napolitano, Julien Le Kernec

**Fake It Till Make It: Federated Learning with Consensus-Oriented Generation** (ICLR2024) Dec 2023
Rui Ye, ***Yaxin Du***, Zhenyang Ni, Siheng Chen, Yanfeng Wang

**Radar-based Human Activity Classification with Cyclostationarity** (CIE Radar 2021) Oct 2021
***Yaxin Du***, Jipeng Li, Zhouyixian Li, Ran Yu, Antonio Napolitano, Francesco Fioranelli, Julien Le Kernec

## Awards

National Scholarship (China) Dec. 2019
Excellent Graduate of Sichuan Province Jun. 2022
Outstanding Student of the College (Top 5/526) Dec. 2021

## Skills

**Programming:** Python, C++, PyTorch
**English:** CET-6 (619), IELTS (7.0)